

[Biagolini et al. 2017](#)

Supplemental Table 1

Original file format: .docx

Copied table to blank .xlsx file

Reordered columns, renamed duplicate columns, find-replaced “ “ with “ _ ” in species names, find-replaced blank cells with “NA”

Deleted extra lines in 2-line species

Saved as “Biagolini_SupTable1.csv”

No additional alterations necessary

28 non-matching species - replaced, then removed duplicates and non-matches species

File name: "Biagolini_data_BirdTreeNames_nodups.csv"

Table caption: Updated review on articles revealing avian extra-pair paternity data. Order (OR):

1=Accipitriformes, 2=Anseriformes, 3=Bucerotiformes, 4=Caprimulgiformes, 5=Cathartiformes, 6=Charadriiformes, 7=Coraciiformes, 8=Cuculiformes, 9=Falconiformes, 10=Galliformes, 11=Gaviiformes, 12=Gruiformes, 13=Mesitornithiformes, 14=Passeriformes, 15=Piciformes, 16=Procellariiformes, 17=Psittaciformes, 18=Sphenisciformes, 19=Strigiformes, 20=Struthioniformes, 21=Suliformes; Breeding System (BS): SM=Social Monogamy, PG=Polygyny, PA=Polyandry, PO=Polygynandry, CB=Cooperative Breeder, SA= Serially Polyandrous, NS= No Social Association; EPP1 (percentage of offspring sired by a partner that is not the social parent); EPB1 (percentage of broods containing at least one extra-pair young); EPP2 (percentage of offspring sired by a partner that is not the dominant in the social group); EPB2 (percentage of broods containing at least one young sired by a partner that is not the dominant in the social group), and n = number of sampled offspring for EPP, and number of sampled broods for EPB.; References (Ref); NA = Not Available.

[Cockburn 2006](#)

Original file format: .pdf

Run code “extract_convert_data_Cockburn_2006.R” with .pdf file as input

Cleaned data outputted as .csv

No additional alterations necessary (“Cockburn2006_data.csv”)

Replaced unmatched species, deleted duplicated entries, deleted entries with no BirdTree equivalent

File name: "Cockburn2006_data_BirdTreeNames_nodups.csv"

[Downing et al. 2015](#)

Original file format: .xls (two tabs)

Saved Table 2 (second tab) as “Downing supp table2.csv”

Several species have multiple lines to report multiple sources for promiscuity or survival%, species name only listed on first line

In Excel, sorted by species and deleted lines that didn’t have “species” listed, saved as “Downing supp table2_kts edited.csv”

No additional alterations necessary

All 238 species in BirdTree

* only species classified as "kinCooperator" were treated as cooperative in our analyses

[Dunn et al. 2015](#)

Original file format: .csv (Data977.csv)
Checked that all species names were present in BirdTree
Resaved as "Dunn supp data 977sp.csv"
No alterations necessary
All 977 species in BirdTree

[Jetz and Rubenstein 2011](#)

Table S2
Original file format: .pdf
Run code "extract_convert_data_Jetz_2011.R" with .pdf file as input
Cleaned data outputted as .csv
No additional alterations necessary ("Jetz2011_data_clean.csv")
Replaced species that don't match BirdTree by hand in Excel
Deleted duplicate entries (whichever species was less similar to the one in BirdTree) and entries without associated species in BT
File name: "Jetz_data_BirdTreeNames_nodups.csv"

Table caption: Information on passerine species considered in this study. Listed are Group (Non-Passerine [NP], Passerine [P]) and Clade membership (Clade), main biogeographic realm (Realm; AT - Afrotropics, PA - Palearctic, NA - Nearctic, NT - Neotropics, IM - Indomalay, AA - Australasia, OC - Oceania), binary breeding system (System), and sources used for cooperative breeding information (see References in Supplemental Information).

[Riehl 2013](#)

Original file format: .pdf
Run code "extract_convert_data_Riehl_2013.R" with .pdf file as input
Outputs .csv file ("Riehl 2013 supp data columns lines.csv") (very messy file)
Edited format in Excel - removed blank cells, combined cells that extend over two lines ("Riehl 2013 supp data columns lines_kts edited.csv" - from summer 2020)
Two species had 2 populations (Melanerpes_formicivorus, Porphyrio_porphyrus): combined data from each column where populations have different values, separated by ";" in Excel
Acrocephalus_vaughani_taiti → Acrocephalus_vaughani in Excel
No additional alterations necessary ("Riehl 2013 supp data columns lines_kts edited.csv")

Supplementary Table 1. Social breeding system and kin structure of breeding group for cooperatively breeding birds. Classifications are as follows:

Column 2: Primary pattern of dispersal that gives rise to cooperative groups. D = delayed dispersal (offspring remain on natal territory after fledging); I = immigration (unrelated individual joins existing breeding group); C = coalition (several same-sex individuals form a group to establish a breeding site or territory).

Column 3: Frequency of cooperative breeding by non-kin, given as the percentage of helpers in the population that are unrelated to the breeding pair (for helper-at-the-nest systems); or as the percentage of adult group members in the population who are unrelated to any other group member (for cooperatively polygamous species). "Majority" indicates that non-kin cooperation is the typical breeding system; "frequent" indicates that exact percentages were not given but that the available data suggests that non-kin cooperation is common in the population; a long dash (—) indicates that kin-based cooperation is the typical breeding system and that unrelated immigrants account for fewer than 20% of adult group members or helpers.

Column 4: Type of data used to calculate kin relationships in the social group. B = behavioural observations (for kin groups only,

including observations of recently fledged juveniles feeding apparent younger siblings; or when authors suspected helpers to be non-dispersed offspring without evidence from marked individuals); R = colour-ringing data, based on known genealogy of marked individuals ; G = genetic data, from microsatellite genotyping, DNA fingerprinting, or protein electrophoresis.

Column 5: Frequency of cooperative breeding in the population. O = obligate (>95% of breeding units are cooperative groups); F = facultative (>5% of breeding units are lone pairs). "Obligate" is used to indicate that nesting by lone pairs was rare in the population studied, not that the species has necessarily lost solitary breeding entirely.

Column 6: Kin structure of social group: primarily composed of kin (K); composed of non-kin (NK); or mixed (M, regularly contain both related and unrelated helpers or co-breeding adult group members).

Column 7: Social breeding system, when cooperative; genetic mating patterns are also given when it is known that they deviate from the social mating pattern.

Column 8: Classification of social breeding system: 1 = pair-nesting with related helpers; 2 = pair-nesting with unrelated helpers or with both related and unrelated helpers; 3 = cooperative polygamy or joint-nesting by non-relatives; 4 = cooperative polygamy or joint-nesting by relatives and non-relatives, with related or unrelated helpers.

[Rubenstein and Lovette 2007](#)

Original file format: .pdf

Run `extract_convert_data_RubensteinLovette2007.R`

Outputs "RubensteinLovette2007_data_clean.csv"

No additional alteration necessary

"RubensteinLovette_data_BirdTreeNames.csv"

Social systems were divided into cooperative and noncooperative. Habitats were divided into savanna and nonsavanna (i.e. desert and forest).

Two species known to show facultative cooperative breeding (*C. leucogaster* and *L. hildebrandti*) were classified as cooperative in our analyses. Mean values of Colwell's measures of predictability (P), constancy (C), and contingency (M) for each species are also listed.

^a D.R.R. personal observation.

^b This species might also inhabit forest-savanna mosaics, but we classify its primary habitat as savanna.

^c This species inhabits thickets and riverine and coastal forests, which we classify as forest.

^d These species are typically found in arid, rocky highland habitats, which we classify as desert.

^e Because this species is associated with riverine gorges, which are usually, but not always, in forests, we classify its primary habitat as forest.

[Griesser and Suzuki 2016](#)

Copied text from Supp Table 1 .docx into Excel

Saved as "Griesser supp table1.csv"

No alterations necessary

All 152 species in BirdTree

Supplement Table 1. List of species that are categorized as occasional cooperatively breeding species (sorted by taxonomic family), following Cockburn [1], and including Darwin finches *Geospiza scandens* and *G. fortis* based on a detailed study on helping at the nest in these two species [2]. Taxonomy follows Jetz et al. [3], parental care mode and the time offspring remain with their parents beyond independency (family time) follows Drobniak et al. [4]. Non-family-living species where offspring disperse soon after independency (no_fam), family-living species where offspring remain well beyond independency with their parents (family), cooperatively breeding, family-living species where offspring remain with their parents and help raising younger siblings (coop_family). Number of Zoological Record entries are obtained from Valcu et al. [5].

[Griesser et al. 2017](#)

Data has tricky species name pattern - common name, last word first

Used eBird common names to match their naming pattern, matched with eBird scientific names, checked BirdTree for those names. If not present, searched manually.